



Secure sharing of industrial IoT data based on distributed trust management and trusted execution environments: a federated learning approach

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Abstract

Industrial Internet of Things (I-IoT) has become an emerging driver to operate industrial systems and a primary empowerer to future industries. With the advanced technologies such as artificial intelligence (AI) and machine learning widely used in IoT, the Industrial IoT is also witnessing changes driven by new technologies. Generally, AI technologies require centralized data collection and processing to learn from the data to obtain viable models for application. In industrial IoT, data security and privacy problems associated with reliable and interconnected end devices are being faced and reliable solutions are urgently needed. A trusted execution environment in IoT devices is gradually becoming a feasible approach, and a distributed solution is a natural choice for artificial intelligence technologies in I-IoT. Moreover, Federated Learning as a distributed machine learning paradigm with privacy-preserving properties can be used in I-IoT. This paper introduces a feasible secure data circulation and sharing scheme for I-IoT devices in a trusted implementation platform by employing federated learning. The suggested framework has proved to be efficient, reliable, and accurate.

Keywords Industrial internet of things · Trusted execution environment · Federated learning · Data security sharing

1 Introduction

The Industrial Internet of Things (I-IoT) is an emerging service-driven industrial ecosystem. I-IoT realizes the effective utilization of resources by interconnecting networks, flowing data, devising a system interoperating industrial resources, optimizing processes for manufacturing, and adapting rapidly to a manufacturing environment.

In addition to the I-IoT, it has been used as an extension of IoT for interconnecting numerous industrial devices, analysis departments, and staff. I-IoT enables the automated manufacture of various applications in various scenarios through a harmonious collaboration of mechanical devices and ultimately achieves Industry 4.0 [1].

I-IoT is characterized by intelligent perception, universal connectivity, exact control, digital modeling, analysis based on real-time, and iterative optimization. I-IoT contains several instruments and generates huge data. Effective data sharing can improve the service quality of emerging applications. Nevertheless, issues pertinent to security and privacy, for example, data leakage have been significant barriers that limit the sharing of data in the I-IoT [2]. Severe consequences can occur for the provider as well as financial losses if protected data were accessed by outsiders. Thus, protecting the privacy and providing high data security in the I-IoT have been a main challenge and require feasible security techniques to form a reliable paradigm for sharing data applications while ensuring data security and privacy.

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To address the security issues in I-IoT, TEE has received much attention, and it is an effective solution [3] that can be well migrated from typical IoT scenarios to I-IoT. TEE is an isolated implementation platform on an instrument independent of an untrusted operating system. TEE furnishes both secure and confidential space to protect data and conduct sensitive computations in untrusted environments, and their security relies generally on hardware-related mechanisms [4]. Since IoT end devices vary greatly in form and function, both general intelligent end devices and simple functional end devices need to be handled differently. For general intelligent end devices, they usually have high hardware configuration, large storage space, operating system, and support multiple network access methods. Therefore, trusted computing modules can be embedded to realize complete trusted computing functions. In addition, these devices further enhance their security capabilities by combining technologies such as secure computing. In I-IoT, applying TEE and other trusted computing technologies to end devices is an effective means of security protection.

For I-IoT scenarios, large-scale device interconnection can effectively improve overall system efficiency. Industrial resources are interconnected by wired or wireless means to form a convenient and efficient industrial IoT information channel. These industrial resources' data interconnection expands the breadth and depth of machine-to-machine, machine-to-human, and machine-to-environment connections. The interconnection of IoT instruments generates a huge quantity of heterogeneous data. Sharing of data helps to build a better application paradigm, but an effective approach is needed for data security and privacy issues. FL [5] emerges as a collaborative AI approach that could function as AI training at the edge of the network. It can coordinate multiple I-IoT devices and machines and it can protect users' privacy. It is particularly attractive for smart I-IoT networks, and it is an effective solution to address data security and privacy and to ensure efficiency in the I-IoT [6]. In addition, edge scenarios are diverse, with many different participating devices. The presence of malicious devices and the trustworthy behavior of end devices involved in federation learning are key to the security of the whole system. It is crucial to ensure better system efficiency and overall performance.

This paper introduces the application of IoT devices under a trusted I-IoT execution environment. Moreover, it proposes a fusion of trusted computing in I-IoT scenarios, to solve the problems related to both security and trustworthiness concerning IoT device terminals. Then, to resolve the potential data security and privacy problems caused by data sharing and circulation in I-IoT interconnection, a federated learning approach is introduced as a distributed AI model capable to improve overall efficiency

by ensuring data security and privacy. By introducing privacy-preserving federated learning, the problem called data-sharing was converted into the problem of a machine-learning. The data privacy was kept well by sharing the parameters of the model instead of leaking the real data. Eventually, an FL approach based on trusted IoT devices is proposed to achieve data security in large-scale industrial applications. The experiment indicated that the suggested scheme was found to be effective.

2 Related work

2.1 Federated learning

FL [7] emerges as a machine-learning paradigm having been extensively studied by researchers. FL is dedicated to addressing data privacy issues in machine learning models or artificial intelligence applications. Mothukuri et al. proposed a series of approaches to protect customer's private data, from privacy-preserving algorithms and joint learning and training methods [8]. The heterogeneity characteristics, statistical heterogeneity, and system heterogeneity of federated learning systems [9] will affect model performance, model convergence speed, and communication efficiency and overhead. How to effectively reduce the impact of heterogeneity is a key challenge. A Federated Averaging Algorithm (FedAvg) is proposed, and it is experimentally demonstrated that it is an efficient communication framework. It converges in the statistical heterogeneity setting, but there is still room for accuracy improvement. Using the FL approach in the industrial IoT, each IoT instrument could train its model by utilizing the data collected locally. The local dataset collected by the IoT instruments has no need to be sent to the centralized cloud that merely needs to collect the updated local training models from individual users. Several implementations adopt FL since it has useful features, e.g., FL is used to enhance suggestions related to Google keyboard [10], healthcare [11], smart city sensing [12], and medicine [13]. Figure 1 depicts an overview of FL in the system of the IoT.

Typically, each IoT instrument has its data and the server aggregating information is situated at the edge of the network or in a virtual cloud of a remotely located cloud computing system [14]. Each FL network has its pros and cons relying on different features. For instance, FL having servers located at the edge of the network is capable of implementations that require low latency, location awareness, and information of network context [15], while FL based on cloud is capable of implementations having many IoT instruments in multiple regions and the requirement for powerful computation/storage capacity.

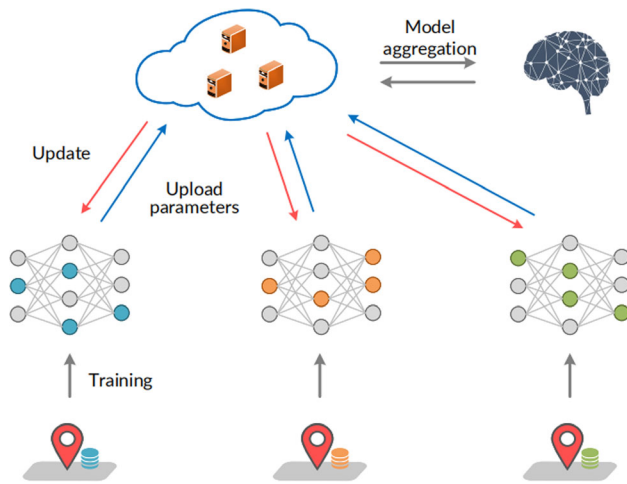


Fig. 1 An illustration of federated learning

Moreover, while FL provided several advantages in enabling learning solutions related to preserving privacy for IoT systems, substantial obstacles such as resource management, robustness, security, and incentives were also observed. In I-IoT, which requires high security, safety, and reliability, these challenges will become even more significant and difficult to address. To make FL work in I-IoT implementations, many investigations have been done. Lu et al., for instance, studied data-sharing implementations of I-IoT by utilizing blockchain and FL. Their experimental results indicated that the suggested framework could improve security and performance utility with no need to train a central server [16]. Both blockchain and FL were employed to pinpoint failures of devices in I-IoT platforms and to protect the raw data of I-IoT instruments [17]. An FL method having communication efficiency was proposed to pinpoint faults occurring in devices of I-IoT by using FL and a long-short-term memory approach [18]. The preservation of data privacy through FL was realized as the advantage of the implementation, while the data characterized by time series analysis in I-IoT could be effectively managed through long and short-term memory approaches. Pham et al. provided a detailed summary of the research on federal learning in industrial IoT, covering all aspects of FL and I-IoT [19].

2.2 Trusted execution environment (TEE)

TEE is intended to enhance the priorly delineated Trusted Platform and is typically referred to as an isolated processing platform in which implementations could be safely performed without regard to the remaining of the system. Nonetheless, TEE still deprives of an exact description and characterizing building blocks that would allow its design to be systematized. As modern systems become more

complex, open, and interconnected, new challenges emerge and traditional security techniques do not provide the security needs of such configurations. This illustrates the latest tendency of incorporating trusted computational concepts into distinct systems (e.g., embedded systems) [20].

Trusted computation is delineated as helping systems to realize secure computing, protecting both privacy and data. Initially, trusted computation relied on a different hardware module that provided a functional interface for the security of the platform. The Trusted Platform Module (TPM) [21] lets systems give evidence of their integrity and safeguard the keys within the tamper-proof hardware module. However, TPM has a primary deficiency that does not allow third parties to have an isolated platform to conduct executions, thus decreasing its functionality to a set of pre-defined APIs. A new method to the trusted computing problem is to let arbitrary execution in a restricted platform that gives tamper-proof execution for its implementations. A TEE is a secure, integrity-preserved processing platform composed of memory and storage capacity [22]. The abbreviation, TEE, is widely utilized when advertised by chip vendors and platform providers for marketing [23]. The initial system deploying TEE was marketed about 10 years ago and was a joint venture of Orange, Trusted Logic, and STMicroelectronics. However, to date, no common and accurate comprehension of the terminology has been constructed, and no framework for evaluating and comparing TEE solutions has been proposed.

2.3 Distributed trust management (DTM)

The distributed trust management module is deployed at the I-IoT edge server. Trust metrics can be measured, calculated, maintained, and shared based on device-to-device behaviors, and stored in a blockchain. This approach ensures a unique trust and enables device trust profiles to be shared with multiple edge servers. Taking the region (cluster) of a single edge server as an example, the definition of trust in this paper is given, including local trust, feedback trust, and global trust. Suppose there are N end devices in the region that have interaction with the edge server.

1. Local Trust (LT). It refers to the trust relationship that exists between two devices, which can be evaluated by the behavior between the two devices during the interaction using local trust metrics. Assuming that there is an interaction between device i and device j , the local trust obtained after the local trust metric can be represented as LT_{ij} .

2. Feedback Trust (FT). The device i periodically trusts feedback to the edge server, i.e., its local trusts of all other

devices in the region form a trust feedback list $FT_i = LT_{i1}, LT_{i2}, \dots, LT_{iN}$. When all devices in the region submit the trust feedback list to the edge server, the trust feedback matrix FM^t can be formed as follows.

$$FM^t = [FT_1^t, FT_2^t, \dots, FT_n^t]^T = \begin{bmatrix} LT_{11}^t & \dots & LT_{n1}^t \\ \vdots & \ddots & \vdots \\ LT_{1n}^t & \dots & LT_{nn}^t \end{bmatrix} \quad (1)$$

3. Global Trust (GT). The device i has a unique global trust in the region it belongs to, which can be represented by GT_i . GT can be calculated by utilizing the feedback trust list of the instrument with the trust feedback matrix. GT is also used to filter whether a device is trustworthy or not key.

3 Overall architecture of proposed model

Federal learning can be well integrated with I-IoT to facilitate end-device collaboration, but still has some issues and challenges. There are many interactions between the end instruments of I-IoT. The behavior of the devices cannot be directly discerned, so there may be malicious devices or devices with malicious behavior. If a malicious device is involved in the training of the federation learning, it will cause serious damage to the model. It may change the optimization direction and performance of the global model, making the trained model unusable. Therefore, a distributed trust management-based I-IoT federated learning model has been proposed for the end-device trust and system performance problems, as depicted in Fig. 2.

The distributed trust management module can calculate and update the trust score of instruments based on the interaction behavior between devices using the defined trust metric. In turn, malicious devices that do not meet the trust requirements are filtered and only trustable devices are selected to participate in federation learning training. The distributed trust management module in the model effectively improves the security of the federation learning training process in the I-IoT. Thus, based on the filtered trusted participating devices, the federation learning training and optimization process can be coordinated by the edge server.

4 I-IoT Trusted execution environment

4.1 Security of IoT devices

I-IoT application system generally includes three parts: IoT end devices, communication network, and IoT server. Due to the huge quantity and diversity of IoT end instruments,

covering various fields and penetrating various industries, as well as the low-security protection capability, most of the attacks are launched from I-IoT end devices. The possible security issues of IoT device end devices are summarized as follows:

1. For I-IoT, end devices usually have limited resources. It is difficult to implement complex security measures due to cost, size, and power consumption.
2. The long life cycle of IoT end devices makes them vulnerable to attacks, which is the main hidden danger. Some security measures with proactive protection capabilities need to be considered.
3. I-IoT end device users and program developers lack security awareness and they often use insecure system configurations. This is a common problem with I-IoT end devices.
4. Some IoT end devices are damaged and access the network or servers without a security assessment. This case poses a security risk to the entire system and requires a security assessment and access control.

IoT end devices are the focus of IoT security and include intelligent end devices and simple functional end devices. For specific implementations, IoT end devices need to be handled in different ways. For the former, they often have high-level hardware, large storage space, operating systems, and support multiple network access methods. As a result, they can embed trusted computing modules to achieve full trusted computing functionality. And they can further enhance their security protection capabilities by combining other technologies. For the latter, building an end-to-end secure and reliable IoT system requires the implementation of security protection measures in the end devices themselves, combined with trusted authentication and corresponding services.

The security of industrial IoT devices requires reliable security management. Combined with the IoT service platform, it enables trusted access and full lifecycle management. In the article, a distributed trust management approach was proposed by utilizing different industrial IoT devices and the trusted execution environment.

4.2 Application of trusted execution environment

TEE is a trusted, isolated, and independent implementation platform on an instrument that exists independently of an untrusted operating system. Both secure and confidential spaces to deal with private data and conduct sensitive computations in untrusted environments can be realized, and hardware-related mechanisms are generally responsible for their security. TEE includes Intel Software Guard Extensions (SGX) [24], Intel management engine

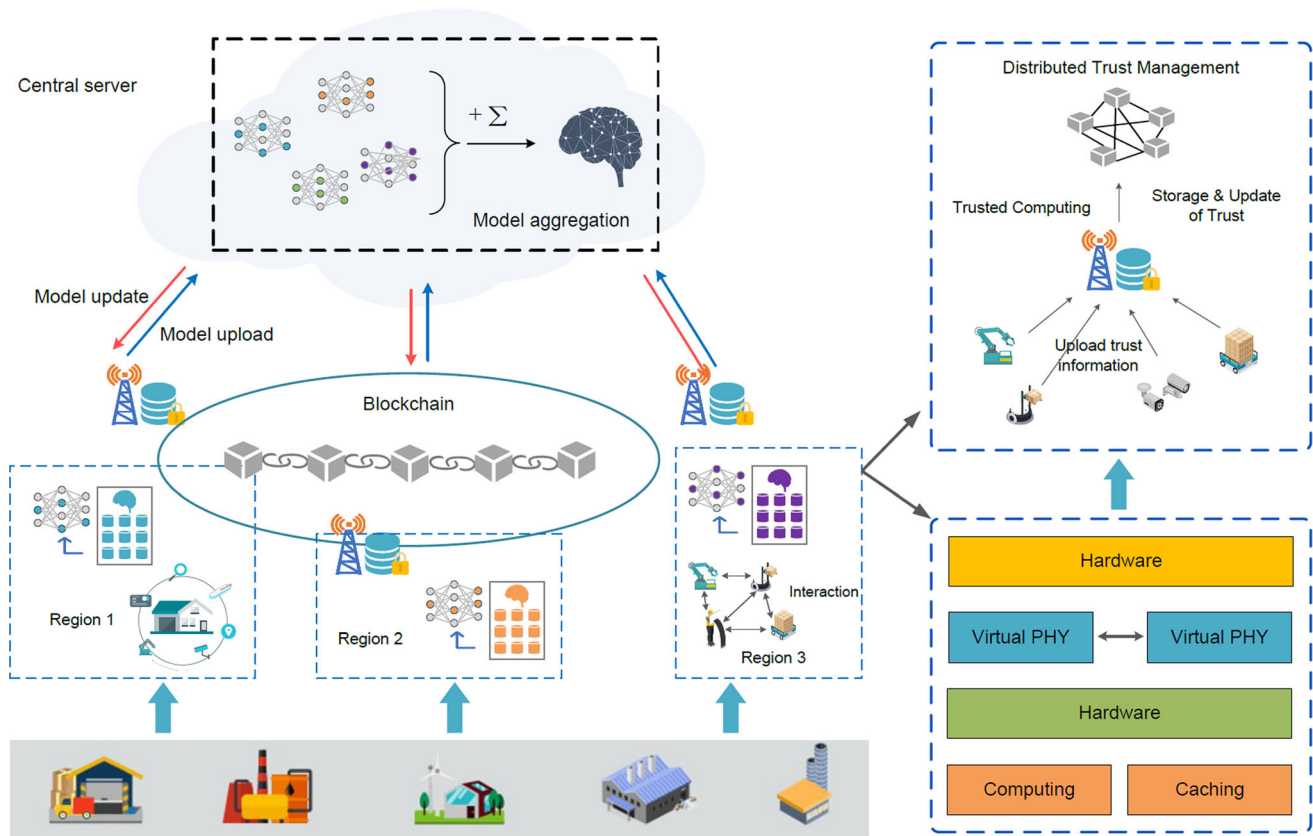


Fig. 2 Illustration of federated learning

(management engine, ME) [25], × 86 system management mode (system management mode, SMM)[26], and AMD memory encryption[27], AMD platform security processor (PSP) [28], and ARM Trust-Zone technology [29].

TEE is commonly used for operations with high-level security requirements, protecting sensitive data, and protecting high-value data. The main purpose of IoT devices with trusted execution environments is to form trusted and reliable end devices. In I-IoT, there is a high demand for security and trustworthiness. In case of a malicious attack, it may cause a huge economic loss. Therefore, the application of TEE in IoT is of great importance.

For the diverse end devices with different computing capabilities in I-IoT, this paper employs TEE for the end devices with high performance that are focused on protection. For other devices, we design a distributed trusted authentication mechanism to assure that the instruments involved in the training of the model are trusted. Overall, this model ensures the data security and privacy of devices when applied in industrial IoT scenarios. In the subsequent section, the distributed trusted authentication approach will be described in detail.

5 Distributed trust management for devices

The distributed trust management module is deployed in the I-IoT edge server, which metrics and updates the trust of end devices. These trust values are stored in a blockchain to ensure uniqueness and immutability. DTM mainly consists of two sub-modules, trust feedback, and global trust. The following is an example of trust management within a single region, and the main components of DTM are introduced.

The trust feedback module is based on the local trust of devices and processes the feedback trust data collected from different end devices. Using the local trust metric, the local trust of device i in a region to other devices is calculated. These trust values can generate a local trust list, which in turn constructs a feedback trust list and a trust feedback matrix. The global trust module is mainly used to calculate and update the global trust of all instruments in each region. The trust list of feedback submitted by all devices is periodically queried at a set time interval. And a global trust list including the global trust of the whole instruments in the region is calculated according to the global trust metric.

After the global trust calculation is completed, a trust update request can be sent to the deployed blockchain layer

to update the global trust list for maintaining the region. All trust storage and updates in DTM can be performed on a blockchain. That is, an all-accessible blockchain layer is jointly deployed among all edge servers, with end devices as client nodes and edge servers as master nodes participating in blockchain consensus.

5.1 Local trust metric

When measuring the trust of various devices, their interaction behavior needs to be carefully considered. Each device trust has unique characteristics, and this section provides a detailed description of the local trust metrics and computation methods.

When device i and device j interact within the time interval Δt device i can calculate the local trust LT_{ij} of device i to device j based on the behavioral performance of device j . Different types of devices provide different services and interactions, so the trust properties that a device has must be taken into account when measuring the device's trust profile. Trust attributes are defined based on the services provided by the device. For example, intelligent devices provide data transfer and computation offloading services, and sensing devices provide data collection services. These devices have different trust attributes among themselves. Therefore, a Trust Attribute Matrix (TAM) can be defined for computing local trust based on characteristics such as the type of device and the type of service provided, as follows:

$$\text{TAM} = \begin{bmatrix} A_{11} & \cdots & \cdots \\ \vdots & \ddots & A_{pq} \\ \cdots & \cdots & \cdots \end{bmatrix} \quad (2)$$

where A_{pq} denotes the trust attribute corresponding to the class q service provided by the class p device. When a device needs a local trust metric for its interaction objects, the appropriate trust attribute can be selected from the TAM based on the service type. Suppose the set of the selected trustset features $A = \{a_1, a_2, \dots, a_g, \dots, a_m\}$, where a_g denotes the g th trust feature. The trust corresponds to a trust feature that was defined as the trust component of the local trust that can be obtained by weighing components of multiple trusts. For the trust attribute a_g , there exists a corresponding metric function f_g to make a trust metric for this attribute, and the trust component corresponds to the trust feature a_g that is denoted as $dt_g = f_g(a_g)$.

Based on the trust attribute matrix and the trust component, the local trust between two devices with direct interaction can be calculated. At the time interval t , the local trust calculation between device i and device j can be expressed as:

$$LT_{ij}^t = \sum_{g=1}^m w_g \cdot dt_g \quad (3)$$

where dt_g is the trust component computed by the device i according to the behavior related to the trust feature a_g in this period. w_g is weight in direct interaction, $\sum_{g=1}^m w_g = 1$. Trust attribute weights determine the significance of various trust features in the local trust and can be defined by the system or by the user based on predefined security principles.

In addition, considering that trust is usually time-sensitive, i.e., the historical trust situation also affects the current local trust. Therefore, historical trust needs to be considered when calculating local trust. Assume the historical trust sequence of device i to device j was $\{LT_{ij}^0, LT_{ij}^1, \dots, LT_{ij}^{t-1}\}$. However, the entire historical trust sequence for local computation raises the storage and energy usage of the instrument. Therefore, historical trust data within a dynamic time window can be selected for computing local trust. Suppose the historical trust sequence within a dynamic time window is $\{LT_{ij}^{t-k+1}, \dots, LT_{ij}^{t-1}\}$. The quantity of past trust records contained in this time window is $k-1$. If devices i and j are not directly interrelated in a time slot, i.e., the predefined initial trust can be used to replace the historical trust. Due to the dynamic changes in time, a trust update mechanism can be constructed based on historical trust sequences to calculate local trust. The trust update mechanism constructed in this paper fully considers the time sensitivity. For the current time slot Δt , the local trust update mechanism is

$$LT_{ij}^{t'} = T_h + \lambda^{\Delta T} \cdot \Delta T \quad (4)$$

where T_h is the mean score of past trust within the dynamic window, $\Delta T = LT_{ij}^t - T_h$ is the difference between the local trust and the mean score of past trust, and $0 < \lambda < 1$ is the trust growth factor that controls the growth rate.

Add the local trusts obtained through the trust updating scheme to the historical trust sequence. The trust sequence used to calculate the local trust is $\{LT_{ij}^{t-k+1}, \dots, LT_{ij}^{t'}\}$. Therefore, based on the weighted calculation of the time decay factor w_τ , the final local trust can be calculated by the following

$$LT_{ij}^t = \frac{\sum_{\tau=1}^k w_\tau \cdot LT_{ij}^{t-k+\tau}}{\sum_{\tau=1}^k w_\tau} \quad (5)$$

where $LT_{ij}^{t-k+\tau}$ denotes the τ th historical trust in this dynamic time window, w_τ denotes the time decay factor of the τ th trust record in this time window.

In summary, the calculation of local trust can be performed based on the above equation and the trust decay

factor. With the continuous change of time, the more distant from the current time slot the historical trust possesses less weight, which is fully in line with the characteristics of trust accumulation. Therefore, local trust is mainly influenced by the trust properties of the interactions between devices and the history of trust. By designing the trust update mechanism to take the historical factors into full consideration and considering the decaying nature of trust, we can obtain more accurate trust calculation results.

5.2 Global trust metric

Global trust calculation of IoT devices is based on trust feedback, which needs to aggregate all trust data in one region according to feedback information of instruments to get the global trust of all devices. At each time slot, devices send constructed feedback trust list FT_i to edge servers, which collect feedback trust lists of all devices and form feedback trust matrix FM^t as input to global trust calculation. For security threat scenarios considered in this article, malicious device nodes can collude with one another to improve their indirect trust and thus disrupt the overall IoT quality of service. So devices with malicious feedback behavior need to be filtered to obtain accurate global trust.

We design a malicious feedback filtering mechanism consisting mainly of feedback trust similarity and clustering algorithms. Usually, two normal and honest devices in the network always have similar trust feedback for the same devices, positive feedback for normal instruments, and negative feedback for malicious instruments. Similarly, dishonest feedback makes feedback from malicious instruments to other devices also similar, while normal and malicious devices may have opposite feedback to other instruments. Therefore, whether the feedback behaviors of two nodes are coherent or not can be determined based on feedback similarity between two device nodes. We assume that trust feedback lists submitted to edge servers by device i and device j in t th time slot are FT_i^t and FT_j^t , respectively. The resemblance of feedback between device i and device j is defined by:

$$\text{sim}_{ij}^t = \begin{cases} 1 - \sqrt{\sum_{q \in cs_{ij}^t} w_{ijq}^t \cdot (FT_{iq}^t - FT_{jq}^t)^2}, & |cs_{ij}^t| \neq 0 \\ 0, & \text{otherwise} \end{cases} \quad (6)$$

where cs_{ij}^t denotes a set of instruments having interacted with both device i and device j during t th time slot, FT_{iq}^t and FT_{jq}^t denote feedback trust of device i and device j on device q during time slot Δt , respectively, and w_{ijq}^t denotes difference weight.

$$w_{ijq}^t = \frac{|FT_{iq}^t - FT_{jq}^t|}{\sum_{q \in cs_{ij}^t} |FT_{iq}^t - FT_{jq}^t|} \quad (7)$$

clustering algorithms are utilized to cluster trust feedback data to eliminate the impact of malicious feedback from malicious devices in global trust calculation for distinguishing honest and malicious feedback trust data. By clustering algorithm can effectively filter out honest feedback instruments to participate in the computation of global trust, and feedback trust filtering mechanism by using the similarity of feedback and the K-Means algorithm.

After completing feedback trust filtering, global trust of instruments is obtained by aggregating feedback trust matrix FM_B^t from filtering. In global trust aggregation, different feedback devices are regarded as monitoring indicators. An entropy weighting method can apply to the aggregation process to ensure effective trust quantification and evaluation. For global trust calculation, it first needs to normalize the filtered feedback trust matrix, then compute the entropy value of each feedback node, i.e., the matrix column vector. And then calculate the coefficient of each feedback node in trust aggregation based on entropy score, then eventually calculate the global trust of each instrument based on weight. And final formulation of global trust calculation at time t is

$$GT_i^t = \sum_{j=1}^s w_j \cdot FM_B^t(i, j) \quad (8)$$

where w_j is the weight occupied by feedback nodes.

6 Experimental analysis

Simulation experiments are designed for two security threat scenarios defined to verify the validity of trust measurement and calculation methods. Experiments are implemented on the NetLogo simulation platform [30], a modeling and simulation platform that allows for evolutionary experiments in the system by simulating complex and open system environments using multi-agent ideology. In this section, simulation experiments are performed by simulating interaction behaviors between I-IoT devices for the trust computing model. Experimental parameters are shown in Table 1.

Three experimental scenarios were considered, denoted as Case1, Case2, and Case3, with Case1: NR = 10%, MR = 10% indicating an idle and good network environment, where the number of both cooperative and malicious nodes is 10%, Case2: NR = 20%, MR = 20% indicating a busy and bad network environment, where the number of both cooperative and malicious nodes is 20%, and Case3:

NR = 40%, MR = 40% indicating a highly busy and bad network environment, where the number of both cooperative and malicious nodes is 40%. We validate the efficacy of the proposed trust computation approach against Threat Scenario A in these three scenarios and further validate it against Threat Scenario B. In the verification experiments against Threat Scenario B, the malicious nodes are divided into two parts: MR = 10%/10, MR = 15%/15%, and MR = 20%/20%. For a more intuitive representation of model performance, the recent research literature [31] and the traditional recommendation-based trust model are chosen as comparison schemes to experiment and analyze in the same experimental setting.

As threat scenarios A and B are disruptive to feedback aspects of trust calculation, the overall trust management scheme effectiveness can be evaluated by the task failure rate [32], which can be calculated as

$$FR = \frac{F_{\text{Task}}}{S_{\text{Task}}} \times 100\% \quad (9)$$

where F_{Task} represents task request counts and S_{Task} represents task failure counts.

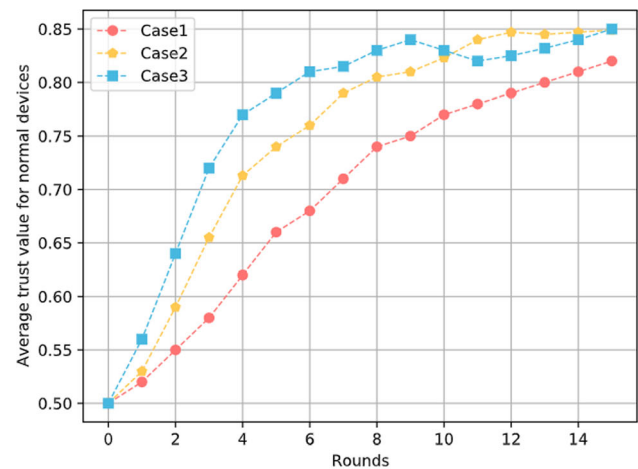
The convergence of the trust calculation scheme is first evaluated by measuring the mean global trust scores of normal instruments. Figure 3 depicts the outcomes of the experiments. These outcomes designate that under threat scenarios A and B, average global trust scores of normal instruments in the different scenarios tend to end up between 0.8 and 0.85 with the increasing number of experimental rounds. It indicates that distributed trust management model achieves better performance in convergence. In our model, trusted devices are selected to participate in FL training based on their global trust, so the experimental results also show that the distributed trust management model enables normal devices to obtain a high global trust, allowing them to be selected to participate in training.

Table 1 Experimental setting for distributed trust management

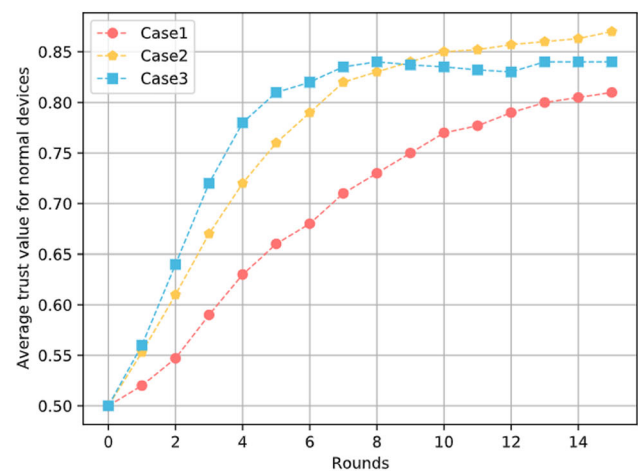
Symbol	Descriptions	Values
n	The number of IoT instruments in a region	50
t	Number of experimental steps	1500/step
Δt	Trust computation time interval	100/step
NR	The rate of cooperative nodes	10/20/40%
MR	The rate of cooperative malicious nodes	10/20/40%
λ_1/λ_2	Trust growth factor	0.2/0.4
γ_1/γ_2	Trust growth threshold	0.3/0.6
α/β	Feedback credibility ratio	0.3/0.7
w	Trust growth threshold	0.7

For task failure rate FR in threat scenario A, experiment results are shown in Fig. 4. From it, we can see that the task failure rate of all schemes in the three cases decreases with time, and the trend is consistent. Although the task failure rate of our scheme in Fig. 4c is slightly higher than the other two comparison schemes, there is not much difference, and it eventually converges. Our proposed scheme is effective in threat scenario A based on FR evaluation metrics.

For treatment scenario B, experiment results are shown in Fig. 5. The task failure rate of the suggested framework was significantly lower than the other two comparison frameworks in all three cases. Over time, the task failure rate decreases for all three schemes, but more significantly in our proposed scheme. Similarly, it is due to considering the impact of collusion attacks when calculating global trust, with filtering and feedback trustworthiness mechanisms to eliminate or reduce such impact to make trust

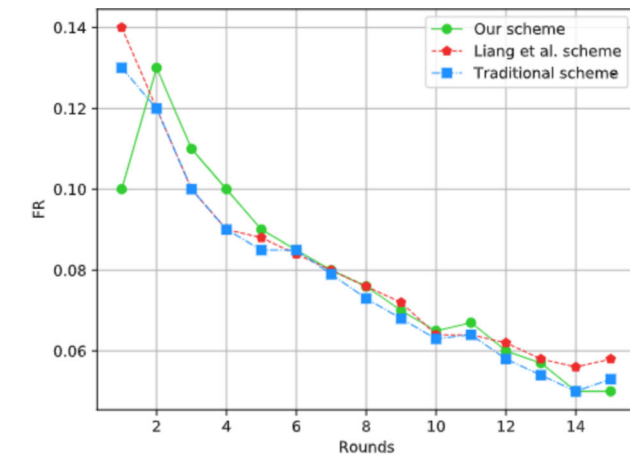


(a) Threat scenario A

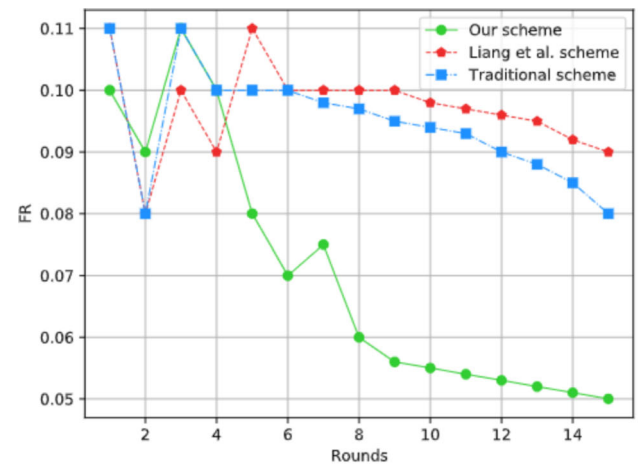


(b) Threat scenario B

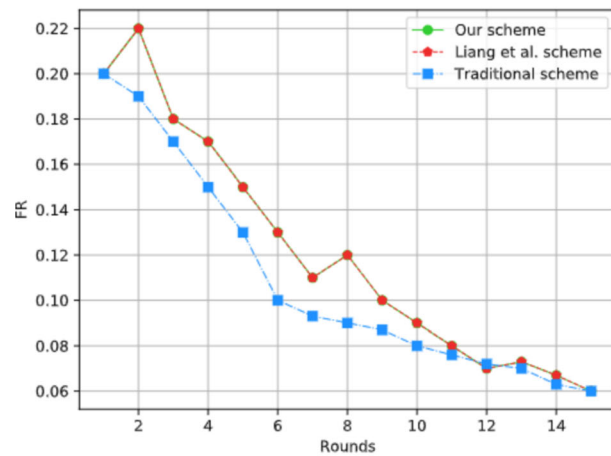
Fig. 3 Average global trust value for normal devices in the different threat setting



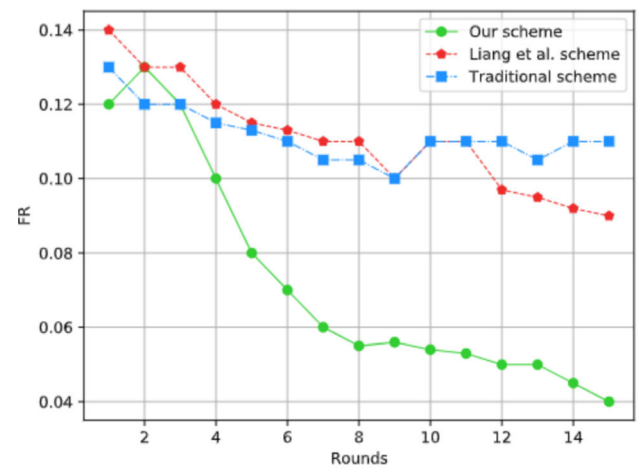
(a) NR=10%, MR=10%



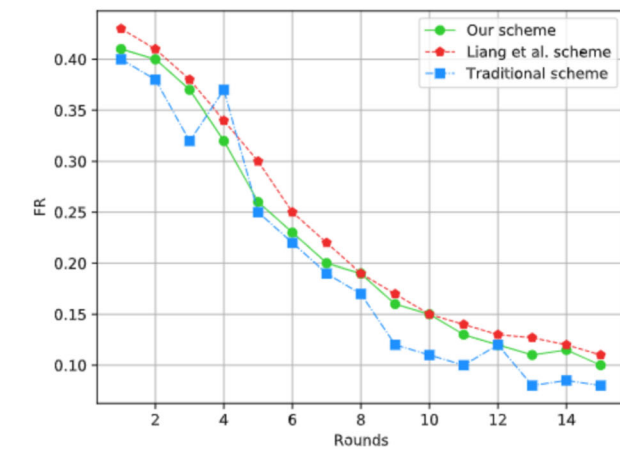
(a) NR=10%, MR=10%/10%



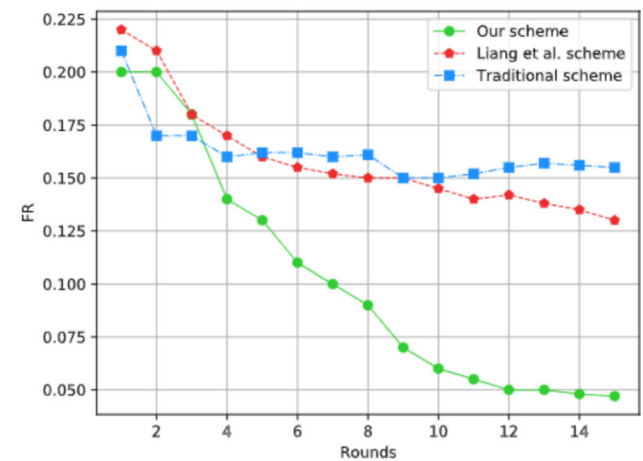
(b) NR=20%, MR=20%



(b) NR=20%, MR=20%/20%



(c) NR=40%, MR=40%



(c) NR=40%, MR=20%/20%

Fig. 4 Task failure rate for the different cases under threat scenario A

Fig. 5 Task failure rate for different cases under threat scenario B

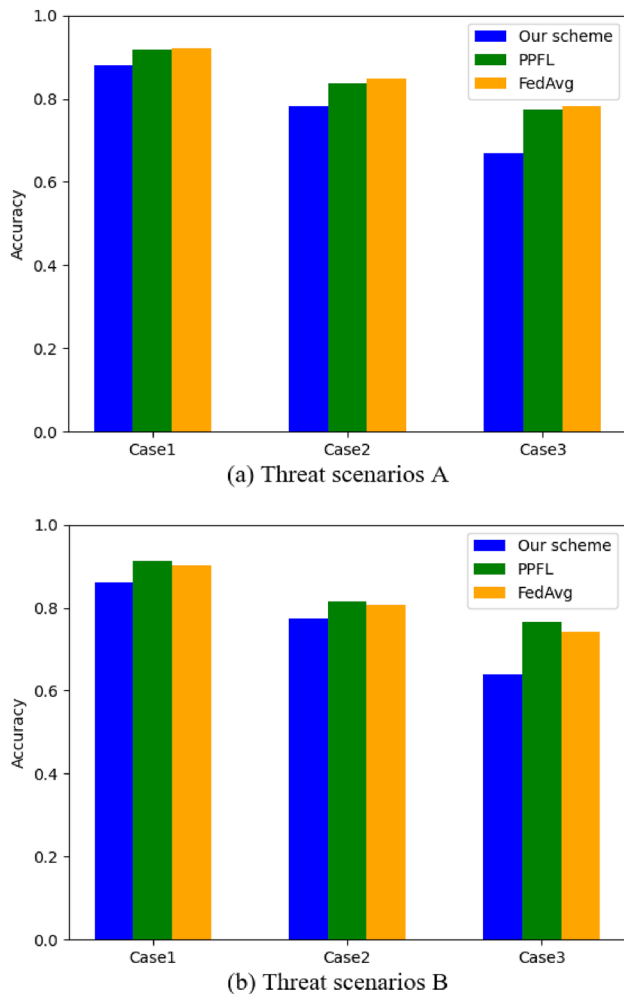


Fig. 6 Accuracy of different models on MNIST in the different scenario

calculation more accurate. Our scheme has better results for threat scenario B based on FR.

In the federated learning experiment, a convolutional neural network (CNN) is utilized for image classification tasks. To accurately evaluate the system performance, we conducted a comparative experiment with FedAvg and PPFL [33] in scenarios A and B. The experimental results are shown in Fig. 6. From the figure, we can see that our model and PPFL outperform FedAvg in terms of accuracy in threat scenario A, which is attributed to the filtering capability of malicious nodes. Although the accuracy of all models decreases with the increase of malicious nodes, our model still exhibits better performance. The results are similar for threat scenario B, but PPFL shows slightly better accuracy than our model in this scenario. Overall, our model is shown to be effective in handling security threats in federated learning.

7 Conclusion

In the article, a federated learning method utilizing a trusted execution environment for IoT devices is proposed for secure data circulation sharing of large-scale IoT device access in I-IoT scenarios. The trusted execution environment is utilized in IoT devices to form trusted devices to ensure secure and trusted data locally in the devices and strengthen the defense capability, and a model for secure data circulation in IoT devices in circulation interconnection is constructed based on the federation learning approach. After experimental validation, our proposed method has better accuracy and efficiency and performs data security and privacy protection effectively.

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Data availability The data implemented in the research can be presented when requested.

Declarations

Conflict of interest The authors declare that they have no conflict of interest.

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